

CS 525: Computational Foundations of Cryptography

Course Code	CS 525
Course Name	Computational Foundations of Cryptography
L-T-P-C	3-0-0-3
Prerequisites	CS 208/CS 513 or equivalent
Intended for	B.Tech / M.Sc / M.Tech / Ph.D
Distribution	All B.Tech / M.Sc / M.Tech / Ph.D
Mutual Exclusion	None
Approval	61st BoA (Jan 2026)

Preamble

This course introduces core cryptographic primitives and demonstrates how their security can be formally established using standard complexity and information-theoretic assumptions.

Course Modules

- **Unit 1:** Introduction to cryptography, Shannon's definitions of perfect secrecy, one-time pad, Shannon's theorem on perfect secrecy. (4 Hours)
- **Unit 2:** One-way functions, equivalence of strong and weak one-way functions, candidate one-way functions, hard-core predicates, Goldreich-Levin theorem. (8 Hours)
- **Unit 3:** Pseudorandom generators (PRG), computational indistinguishability, equivalence of one-way functions and PRGs, construction of PRGs, pseudorandom functions. (10 Hours)
- **Unit 4:** Interactive proofs, interactive proof for graph non-isomorphism, zero-knowledge proofs, zero-knowledge proof for graph isomorphism, zero-knowledge proofs for NP, non-interactive zero-knowledge proofs. (10 Hours)
- **Unit 5:** Introduction to Information Theory, Perfect secrecy and secret-key encryption, Universal hash families, Information reconciliation and privacy amplification, Two-party secure computation with active/passive adversaries, Multiparty secure computation: broadcast, Byzantine agreement. (10 Hours)

Laboratory Modules

None

Text Books

1. Oded Goldreich, *Foundations of Cryptography*, Vol. 1, Cambridge University Press, 2001.

References

1. Salil P. Vadhan, *Pseudorandomness*, Now Publishers, 2012.
2. R. Cramer, I. V. Damgard, and J. B. Nielsen, *Secure Multiparty Computation and Secret Sharing*, Cambridge University Press, 2015.